
SYNCHRONIZATION RECORD OF THE HARDWARE CLOSURE SERIES

*Twenty-Five Engines Recompiled, Twenty-Four Twins Rerun, Six Witnesses Brought to Edition 1.5.0,
and Every Correction of the Series on One Record*

*Sealed at Round Seven of One Adversarial Self-Audit Cycle Across All Six Registers; Every Figure Recomputed
From the Files Themselves*

FOUNDATIONS OF PHYSICS · THE HARDWARE CLOSURE SERIES

Every engine recompiled, every correction on one record

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ABSTRACT. This record synchronizes the hardware closure series with the six witnesses that preceded it. It recompiles twenty-five engines in core Lean 4.19.0 and reruns twenty-four Fortran twins in one session: 1,246 laws, none depending on choice, every one refused under negation, and 286 twin checks with no failure. It brings the six witnesses to edition 1.5.0 with no theorem statement, row status, twin output or grade changed, and records every correction made across the programme.

KEYWORDS *synchronization · census · Lean 4 · Fortran · audit*

1. SCOPE

This record synchronizes the hardware closure series with the six witnesses that preceded it. It covers four acts. - **The witnesses brought forward.** The six witnesses are brought from edition 1.4.0 to edition 1.5.0. - **The census recomputed.** Every engine of the programme is recompiled and every twin rebuilt and rerun, in one session. - **The corrections collected.** Every correction made to a paper of the programme is recorded, whether made after publication, in a draft, or during its build. - **The cross-references mapped.** The witnesses are mapped to the series papers that extend them.

It changes no theorem statement and no row status. It raises no grade.

2. LOAD-BEARING CLAIMS

- **C1.** Every engine of the programme compiles in core Lean 4.19.0 with no axiom declared, and no law of any engine depends on the axiom of choice. *Falsifier:* an engine transcript showing Classical.choice for any law, or any axiom declaration in any engine source.
- **C2.** Every law of every engine was negated in place and refused, and in every judgment a planted vacuous law survived its negation. *Falsifier:* a judgment file whose refused count differs from the engine's law count, or a planted vacuous law refused.
- **C3.** The six witness editions 1.5.0 change no theorem statement, no row status, no twin output and no grade of edition 1.4.0. *Falsifier:* a theorem statement differing between a 1.4.0 and a 1.5.0 witness engine, or a 1.5.0 twin run differing from its published run.
- **C4.** The census totals printed in Section 3 equal the sums of the census rows, and every row equals what the recompilation, the twin's rerun and the engine's own judgment record printed. *Falsifier:* a census total differing from the sum of its rows, or a row differing from a fresh recompilation.
- **C5.** Every falsifier of the series that is bound to a theorem names a theorem declared in the engine the falsifier's paper carries. *Falsifier:* a bound falsifier whose theorem is not declared in the engine its paper carries.

3. THE CENSUS

Every engine was recompiled in core Lean 4.19.0, and every twin rebuilt under the sealed flags `-std=f2018 -02 -fno-fast-math -ffp-contract=off` and rerun, on 1 October 2026. The axiom columns count each law once, by the cone its transcript prints.

Engine	Laws	Axiom-free	Propext	Propext and quotient	Choice	Refused	Twin checks	Engine SHA-256
The Electron Is the Seat, 1.5.0	27	16	1	10	0	27 of 27	17 of 17	ee7ef2a6a09d7e1d
The Neutrino Is the Witness, 1.5.0	12	8	2	2	0	12 of 12	11 of 11	c41c95b6e66a0b0c
The Arrow Has Two Branches, 1.5.0	11	9	0	2	0	11 of 11	9 of 9	de2bfe386f7baeab
The Proton Is the Lock, 1.5.0	11	6	3	2	0	11 of 11	9 of 9	7c2af1a387671bd1
The Magnet Is the Witness of the Pair Correlation, 1.5.0	9	4	0	5	0	9 of 9	6 of 6	c95fe4c6c63ff8f1
The Fluid Is the Witness, 1.5.0	7	2	0	5	0	7 of 7	5 of 5	ab9f2e965d625a5f
One Actuating Substrate (The Four Forces), 2.0.0	80	32	7	41	0	80 of 80	34 of 34	0d6aaa8477662454
The Geometric Nature of Light, 2.0.0	63	27	6	30	0	63 of 63	16 of 16	0734621ab9528352
Electromagnetism71 on the Riemann Locus, 1.0.0		28	6	37	0	71 of 71	17 of 17	bf85af4943dab16c
The Bridges Are One Report	26	8	1	17	0	26 of 26	none	eb4b3492dde91120
Immanent Gravity, 2.0.0	68	27	5	36	0	68 of 68	17 of 17	e5f12bb60a57107f
Spin and Statistics, 1.0.0	69	24	6	39	0	69 of 69	17 of 17	b0de4a65721def7e
The Weak Force Reads the Orientation, 1.0.0	59	24	5	30	0	59 of 59	13 of 13	98ffbd3dd30cd2e3
Resonances Off the Line, 1.0.0	57	17	4	36	0	57 of 57	12 of 12	504dc2664bf023e6
The Ferromagnet Is the Witness of the Circle, 1.0.0	58	21	4	33	0	58 of 58	9 of 9	61af3fe7622be92b
Dark Matter as the Charge Conjugation Fixed Set, 2.0.0	125	42	7	76	0	125 of 125	38 of 38	466a1ef21f0ccca7
The Measurement Is the Registration, 1.0.0	63	21	4	38	0	63 of 63	8 of 8	428af3eb969a19d5
The Arrow of Time Is the Keyed Bit, 1.0.0	54	21	4	29	0	54 of 54	9 of 9	ffadd0599b814013

Engine	Laws	Axiom-free	Propext	Propext and quotient	Choice	Refused	Twin checks	Engine SHA-256
Matter over Antimatter Is the Spent Bit, 1.0.0	54	19	3	32	0	54 of 54	6 of 6	3b5c8e449748a09a
Symmetry Breaking Is the Spent Bit, 1.0.0	56	17	4	35	0	56 of 56	6 of 6	9b554dd7450011e6
The Correlation Lives in the Joint Registration, 1.0.0	55	22	5	28	0	55 of 55	6 of 6	23f2e80448c2b2b7
The Strong Force Reads No Orientation, 1.0.0	48	22	3	23	0	48 of 48	5 of 5	3dff4ca961a1ed9b
The Horizon Is the Registration, 1.0.0	56	22	3	31	0	56 of 56	7 of 7	fb0a1306af373068
What the Seat Cannot Reach, 1.0.0	55	23	3	29	0	55 of 55	6 of 6	5f54f7046d3b6be3
Typed Falsifiability of the Hardware Closure Series, 2.0.0	52	25	3	24	0	52 of 52	3 of 3	c5aaebf94749095a
Total, 25 engines, 24 twins	1246	487	89	670	0	1246 of 1246	286 of 286	

Twenty-five engines carry 1,246 laws. Of these, 487 depend on no axiom, 89 on propositional extensionality alone, and 670 on propositional extensionality and quotient soundness; none depends on choice, and no engine declares an axiom. All 1,246 were refused under negation. Twenty-four twins run 286 checks with no failure. The Bridges report carries an engine and no twin.

Witness	Engine	Change	Laws	Twin
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4. THE SIX WITNESSES, EDITION 1.5.0

Each engine and twin was extracted from edition 1.4.0's appendices and matched byte for byte against its printed SHA-256: six of six engines, six of six twins.

Witness	Engine	Change	Laws	Twin
The Electron Is the Seat	changed	the axiom cones of two unprinted theorems now printed; the printed count corrected from 25 to 27	27	rerun identical
The Neutrino Is the Witness	changed	one unprinted cone now printed; the count 11 of printed correspondences kept, 12 laws stated	12	rerun identical
The Arrow Has Two Branches	changed	magnetism_pair re-proved without choice, statement unchanged; three unprinted cones printed; the printed count corrected from 8 to 11	11	rerun identical
The Proton Is the Lock	byte-identical	none	11	rerun identical
The Magnet Is the Witness of the Pair Correlation	byte-identical	none	9	rerun identical
The Fluid Is the Witness	byte-identical	none	7	rerun identical

The diff of every engine pair shows the added print lines, one final newline, and one proof line of magnetism_pair replaced by two. No theorem statement changed. Each paper gains three things: - a synchronization paragraph naming the series papers that extend it; - the types of its falsifiers in the series ledger; - an edition note stating the changes above.

5. CORRECTIONS ON THE RECORD

Where	Before	After	Found by
The Arrow Has Two Branches	magnetism_pair depended on choice	depends on propositional extensionality and quotient soundness	the sync's recompilation
Electron, Neutrino, Arrow witnesses	printed theorem counts 25, 11 and 8 understated the laws	27, 12 and 11 laws, every cone printed	the sync's recompilation
Spin and Statistics	F-Connection bound to turnSign, a definition	bound to half_integer_spin_is_odd	the ledger twin

Where	Before	After	Found by
Typed Falsifiability, standing five	IV, baryon number exactly conserved	withdrawn; contradicts the theorem that an asymmetry from a symmetric start moves baryon number; replaced by the proton's stability	the ledger's cross-check
Typed Falsifiability, standing five	V, no electron substructure	withdrawn; no engine proves it	the ledger's cross-check
Dark Matter, edition 2.0.0, Section 7	the dark charge zero because nothing actual stands off the line	the dark bit is the coverage premise of the kinetic crossing, spent by the act	the build's own reading
The Correlation Lives in the Joint Registration	the PR box's value stated at the wrong scale	8 on the doubled table, a CHSH value of 4	the engine's refusal
The Strong Force Reads No Orientation	a pre-written susceptibility of 75.9 MeV	the computed 76.19 MeV, labelled computed	the twin's failure

6. CROSS-REFERENCES

Witness	Series papers that extend it
The Electron Is the Seat	Electromagnetism; Spin and Statistics; The Measurement Is the Registration; Dark Matter; Matter over Antimatter
The Neutrino Is the Witness	The Weak Force Reads the Orientation; Dark Matter; Matter over Antimatter; What the Seat Cannot Reach
The Arrow Has Two Branches	Immanent Gravity; The Arrow of Time; The Horizon Is the Registration; Electromagnetism; The Ferromagnet
The Proton Is the Lock	The Strong Force Reads No Orientation; Matter over Antimatter; Symmetry Breaking; Typed Falsifiability
The Magnet Is the Witness of the Pair Correlation	The Ferromagnet; Electromagnetism; Resonances
The Fluid Is the Witness	The Arrow of Time; Resonances

7. THE FALSIFIER LEDGER

The ledger. Typed Falsifiability of the Hardware Closure Series, edition 2.0.0, carries sixty-four falsifiers from twenty-two papers, the six witnesses and the sixteen physics papers of the series. They are typed by target: - eighteen the line; - thirteen the fold; - eleven the wall; - nine positivity; - eight a count; - five the price.

Bound and unbound. Fifty-eight are bound to a theorem and six to an imported theorem or a structural reading. The ledger twin, rerun in this session against the witnesses' 1.5.0 engines and the series' engines as filed, finds all fifty-eight declared where the ledger names them.

The standing five. Of the earlier draft's five standing predictions, three are bound and two withdrawn.

8. INVARIANTS OF THE SYNC

- **No statement changed.** No theorem statement of any witness changed (Section 4).
- **No status or grade changed.** No row status changed. No grade rose: every claim of every witness stands at the grade its edition 1.4.0 printed.
- **Identical twins.** Every witness twin's rerun is byte-identical to its published run.
- **The census.** Every engine of the programme is free of choice and declares no axiom, and every law was refused under negation.

9. OWED

- **Filing.** The single push of every draft, the corrected spin paper, the six witness editions, a regenerated bundle and the index, held for the architect's yes. Then the deposits of the new and revised records.
- **Review.** The external audits of every paper, and the boot on a second substrate that the operating system requires.

10. PROVENANCE

The census was computed by one script from the engines, twins and judgment records themselves, reading each engine's judgment from that engine's own record. Its output and the figures above agree. A first run of the script misread the judgment of eight engines whose records share a folder or a file; the script was corrected and rerun, and the table carries the corrected run.

Audit. This edition passed one adversarial self-audit cycle of seven rounds across all six registers, sealed at round seven. The prosecutions came from the scribe alone. Every arithmetic or record error they found was repaired, six in all, and three errata of the auditing seat were filed against itself, among them an attestation of a re-execution that had not completed, which voided round six's evidence and was repaired by a seventh round run to completion. The planted controls were SELF grade, caught twenty-one of twenty-one; the audit was single-substrate on both sides, and the external review of this record is owed.